AIR UNIVERSITY

DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

EXPERIMENT NO 9

Lab Title Symbolic Toolkit

Student Name: Ali Abbas Khan

Reg. No: 230661

Objective: _____

_____

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks: _____

Date: _____ Signature: _____



Department of Electrical and Computer Engineering

Signal and Systems Lab

Lab Report: 9

Title: Symbolic Toolkit

Name: Ali Abbas Khan

Roll Number: 230661

Class: BCE-5B

Submitted To: Engr. Maryam
Date**:** November 26, 2025

1. Take a vector from x=-5:5 and plot in script file.
Now repeat the same by using Symbolic Mathematics and plot by using ezplot command.

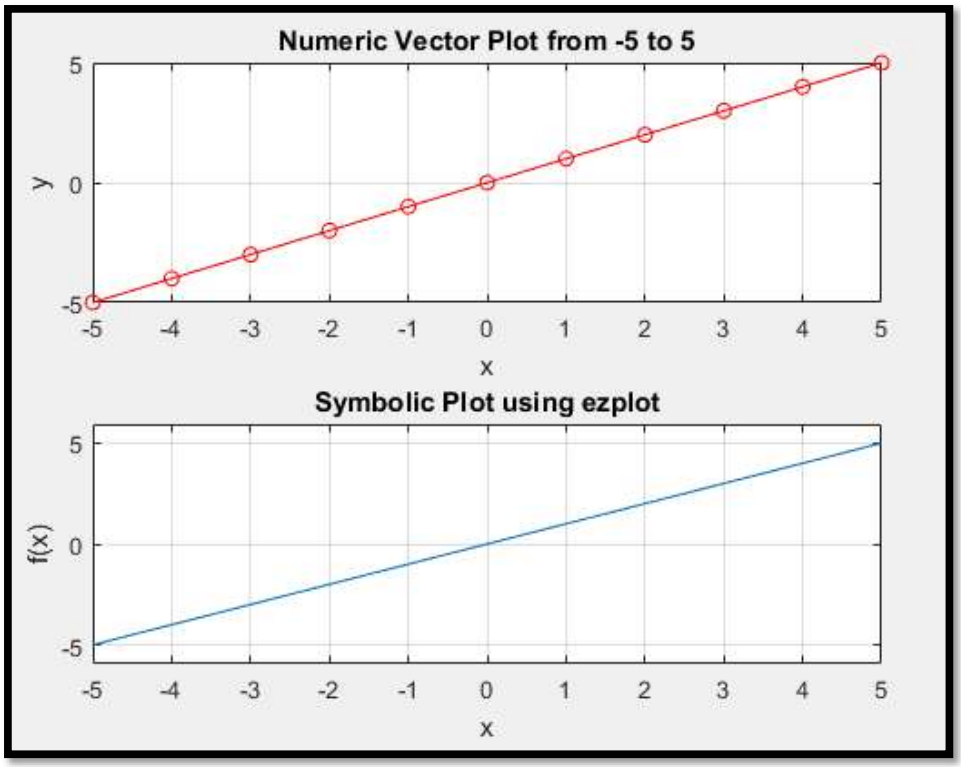
Code:

Editor - C:\Users\Admin\OneDrive\Desktop\SNS Lab\Matlab Files\L9\T1.m

T1.m

```
1 -    x = -5:5;
2 -    y = x;
3 -    figure;
4 -    subplot(2,1,1);
5 -    plot(x, y, 'o-r');
6 -    xlabel('x');
7 -    ylabel('y');
8 -    title('Numeric Vector Plot from -5 to 5');
9 -    grid on;
10
11 -   syms x;
12 -   f = x;
13 -   subplot(2,1,2);
14 -   ezplot(f, [-5, 5]);
15 -   xlabel('x');
16 -   ylabel('f(x)');
17 -   title('Symbolic Plot using ezplot');
18 -   grid on;
```

Output:



Code:

Editor - C:\Users\Admin\OneDrive\Desktop\SNS Lab\Matlab Files\L9\T2.m

T1.m | T2.m

```
syms x y z
eq1 = 6*x - 3*y + 8*z == 10;
eq2 = -9*x + 7*y + 3*z == 2;
S = solve([eq1, eq2], [x, y, z]);
fprintf("Value os X Y and Z\n");
disp(S.x)
disp(S.y)
disp(S.z)
```

Output:

Command Window

```
>> T2
Value os X Y and Z
76/15

34/5

0
```

Code:

Editor - C:\Users\Admin\OneDrive\Desktop\SNS Lab\Matlab Files\L9\T3.m

T1.m | T2.m | T3.m

```
h = 2;
k = -3;
r = 2;

theta = linspace(0, 2*pi, 100);
x_circle = r * cos(theta) + h;
y_circle = r * sin(theta) + k;
plot(x_circle, y_circle, 'b', 'LineWidth', 2);

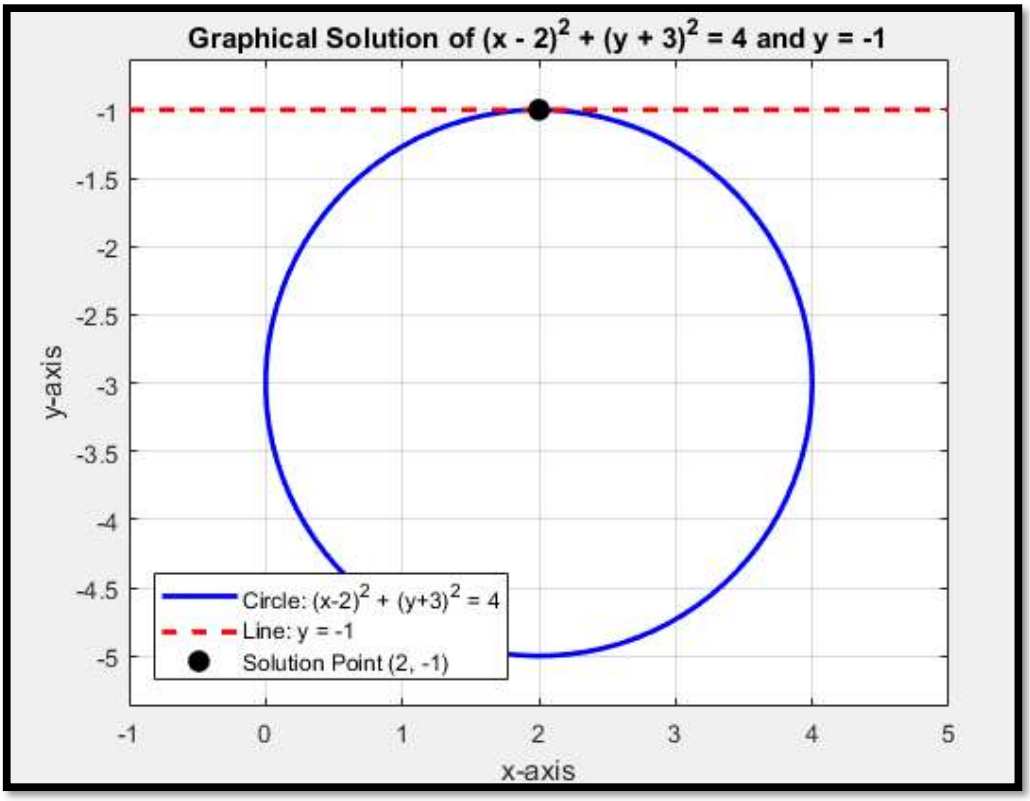
hold on;

x_range = [h - r - 1, h + r + 1];
y_line = [-1, -1];
plot(x_range, y_line, 'r--', 'LineWidth', 2);

intersection_x = 2;
intersection_y = -1;
plot(intersection_x, intersection_y, 'ko', 'MarkerFaceColor', 'k', 'MarkerSize', 8);

title('Graphical Solution of (x - 2)^2 + (y + 3)^2 = 4 and y = -1');
xlabel('x-axis');
ylabel('y-axis');
axis equal;
grid on;
legend('Circle: (x-2)^2 + (y+3)^2 = 4', 'Line: y = -1', 'Solution Point (2, -1)', 'Location', 'southwest');
hold off;
```

Output:



Code:

Editor - C:\Users\Admin\OneDrive\Desktop\SNS Lab\Matlab Files\L9\T4.m

T1.m | T2.m | T3.m | T4.m

```
1 -  syms x
2 -  eq1 = 3*x^2 - 5*x - 4 == 0;
3 -  sol1 = solve(eq1, x);
4 -  fprintf('Solutions for the first equation (3x^2 - 5x - 4 = 0):\n');
5 -  fprintf('x1 = %s\n', char(sol1(1)));
6 -  fprintf('x2 = %s\n', char(sol1(2)));
7 -  fprintf('\n');
8 -  eq2 = 7*x^2 + 10*x - 8 == 0;
9 -  sol2 = solve(eq2, x);
10 - fprintf('Solutions for the second equation (7x^2 + 10x - 8 = 0):\n');
11 - fprintf('x3 = %s\n', char(sol2(1)));
12 - fprintf('x4 = %s\n', char(sol2(2)));
```

Output:

```
>> T4
Solutions for the first equation (3x^2 - 5x - 4 = 0):
x1 = 5/6 - 73^(1/2)/6
x2 = 73^(1/2)/6 + 5/6

Solutions for the second equation (7x^2 + 10x - 8 = 0):
x3 = -2
x4 = 4/7
```

Conclusion:

This lab used graphical and symbolic methods to solve equation systems. We found the intersection point for the line-circle system and confirmed the exact, radical-based solutions for quadratic equations using the symbolic solver. The final analysis confirmed the limitations of the symbolic command, necessitating numerical methods for transcendental problems.